

Experiment 19

Understand Logistic Regression for a given data set.

Program:

```
import numpy as np
import pandas as pd

df=pd.read_csv('Social_Network_Ads.csv')
df
df.head()

features=df.iloc[:, [2,3]].values
label=df.iloc[:,4].values

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

x_train,x_test,y_train,y_test=train_test_split(
    features, label, test_size=0.2, random_state=37
)

model=LogisticRegression()
model.fit(x_train,y_train)

print(model.score(x_train,y_train))
print(model.score(x_test,y_test))

from sklearn.metrics import classification_report
print(classification_report(label,model.predict(features)))
```

Output

Output:
Training score: 0.834375
Testing score: 0.9125

Classification Report:

	precision	recall	f1-score	support
0	0.85	0.93	0.89	257
1	0.84	0.71	0.77	143
accuracy			0.85	400
macro avg	0.85	0.82	0.83	400
weighted avg	0.85	0.85	0.85	400